



Nguyen Khac Minh Tan

Full-Stack Developer

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EDUCATION

B.S. Software Engineer

Van Lang University
2026

CERTS

Olympic PMNM Contest 2025

BTC OLYMPIC TIN HOC SINH VIEN VIET NAM
12/12/2025

ICPC Vietnam Southern Provincial PC

William B.Poucher, Ph.D ICPC Executive
Director 11/10/2025

ICPC Vietnam National PC

William B.Poucher, Ph.D ICPC Executive
Director 8/11/2025

SUMMARY

I build systems that handle real traffic at real scale. Production software that processes millions of transactions, serves millions of users, and stays up when things go wrong. My work sits at the intersection of distributed systems, real-time data pipelines, and platform engineering.

EXPERIENCE

Graduation thesis

2/2026 - 6/2026

Develop a project to monitor and predict traffic density using a microservice system.

Full-Stack Developer — DataFlow Inc

2018 - 2021

Built real-time analytics platforms and streaming data pipelines. Reduced data lag from 24h to under 30s using Flink and Kafka.

PROJECTS

Payment Gateway Migration

2024-01-15

FINTECH

Problem: Legacy monolith processing 2M+ daily transactions with 0.8% failure rate and 12s avg latency.

Results: 99.97% uptime | 180ms p99 latency | 0.01% failure rate

Insight: Idempotency keys are non-negotiable in payment systems.

Tech: Go, Kafka, PostgreSQL, K8s

Real-time Analytics Pipeline

2024-03-22

DATA

Problem: Batch ETL jobs causing 24h data lag, making real-time decisions impossible for ops team.

Results: <30s data lag | 500K events/sec | 40% cost reduction

Insight: Exactly-once is worth the complexity premium.

Tech: Flink, Kafka, ClickHouse, Grafana

Auth Platform Redesign

2024-05-10

SECURITY

Problem: Homegrown auth with 3K LoC, no MFA, and session fixation vulnerabilities across 12 services.

Results: 0 auth vulns | 2M+ daily logins | <50ms token validation

Insight: Security is a property of the system, not a feature.

Tech: Rust, Redis, OAuth2, WebAuthn

E-commerce Platform

2024-06-01

RETAIL

Problem: Black Friday traffic spikes causing cascading failures, \$2M lost revenue per outage minute.

Results: 0 outages | 10x peak traffic | \$18M BF revenue

Insight: Resilience is built by breaking things intentionally first.

Tech: Node.js, Redis, RabbitMQ, AWS

CI/CD Pipeline Overhaul

2024-07-14

DEVOPS

Problem: Build times averaging 45min, flaky tests at 30% failure rate, deploys taking 2h with manual gates.

Results: 4min build time | <1% flaky rate | 15min deploy time

Insight: Fast feedback loops change developer behavior more than any policy.

Tech: Go, K8s, Terraform, ArgoCD

Healthcare Data Platform HEALTHCARE Problem: HIPAA-compliant data silos across 8 hospitals, patient records taking 72h to sync between systems. Results: <5min record sync 100% HIPAA compliance 8 hospitals live Insight: Compliance is architecture, not paperwork. Tech: Rust, PostgreSQL, Kafka, AWS	2024-08-20
Multi-tenant SaaS Platform SAAS Problem: Single-tenant architecture hitting scaling ceiling at 50 customers, onboarding taking 3 weeks per tenant. Results: 500+ tenants 15min onboarding 60% infra cost saved Insight: Multi-tenancy is a data modeling decision, not a deployment strategy. Tech: TypeScript, PostgreSQL, Redis, Docker	2024-09-05
Fraud Detection Engine FINTECH Problem: Rule-based fraud detection missing 35% of fraud cases while generating 90% false positive rate. Results: 92% detection rate <8% false positive <100ms scoring latency Insight: Model accuracy means nothing without operational latency budgets. Tech: Python, Flink, Kafka, Redis	2024-10-12
IoT Fleet Management IOT Problem: 50K+ devices with unreliable connectivity, 40% telemetry data loss, and no remote update capability. Results: 99.5% data delivery 50K+ devices managed <2GB update size Insight: Eventually consistent is the only consistency model that works at the edge. Tech: Go, MQTT, DynamoDB, AWS	2024-11-01
Content Delivery Network INFRA Problem: Origin server handling 100% of traffic, 800ms avg TTFB for APAC users, \$500K/month bandwidth cost. Results: <50ms APAC TTFB 95% cache hit rate \$120K monthly cost Insight: Cache invalidation is only hard if your keys are wrong. Tech: Rust, Nginx, Terraform, AWS	2024-11-18
Search Platform Rebuild DATA Problem: MySQL LIKE queries on 50M records, 12s search latency, zero relevance tuning capability. Results: <80ms search latency 3x CTR improvement 50M documents indexed Insight: Search quality is a product of iteration, not architecture. Tech: Python, Elasticsearch, Kafka, K8s	2024-12-03
Chat & Notification System SAAS Problem: Polling-based messaging consuming 60% of API capacity, 5s message delivery latency, no offline support. Results: <200ms delivery 80% API load reduced 100K concurrent users Insight: Offline-first is not a feature, it is the only sane architecture for mobile. Tech: TypeScript, Redis, WebSocket, CRDTs	2025-01-08
Compliance Automation Platform SECURITY Problem: SOC2 audit requiring 3 months of manual evidence collection, 200+ controls tracked in spreadsheets. Results: 2 weeks audit prep 200+ controls automated 98% evidence coverage	2025-02-14

Insight: Compliance should be a byproduct of good infrastructure, not a separate process.

Tech: Go, Terraform, PostgreSQL, AWS

Video Transcoding Pipeline

2025-03-01

MEDIA

Problem: Manual transcoding workflow, 8h per video, no adaptive bitrate, \$200K/year in compute waste.

Results: 15min per video | 40% storage saved | 4K max resolution

Insight: Per-title encoding pays for itself within a month at scale.

Tech: Go, FFmpeg, AWS, Kafka